Introduction

MAT2040 Linear Algebra

About MAT2040

Instructor: Dongdong He

Lecture schedule:

- ★ L0102: Monday and Wednesday, 10:30am-11:50am, 13:30pm-14:50pm, Teaching Complex C301.

- Office hours:
  - ★ Monday 3pm-5pm.
  - ★ You can also talk to me directly after class, or make the appointment.
- Office: Chengdao Building, Room 313
- E-mail: hedongdong@cuhk.edu.cn

Teaching Assistants:

There are 19 TAs for this course for handling 20 tutorial sessions. Information will posed on the blackboard shortly. This course has totally 4 sessions.

Note: There is no tutorial in the first week.

About MAT2040

Recommended Reading:

- Linear Algebra with Applications (9th Edition)
  by Steven J. Leon.
- Gilbert Strang, Introduction to Linear Algebra, 5th edition, Wellesley-Cambridge Press.
- D.C. Lay, S. R. Lay, J.J. McDonald, Linear Algebra and Its Applications, 5th edition, Pearson.

Course website: Blackboard `https://bb.cuhk.edu.cn/`

Tentative Syllabus

- Linear Systems and Matrices
- Matrix and Matrix Inverse
- Vectors and Vector Spaces
- Determinants
- Linear Transformations
- Orthogonality
- Eigen-Theory
- Quadratic Form and Singular Value Decomposition



Assessment

- Homework-at least 9 assignments (25%)
- Midterm Exam (35%) – Weekend of week 8, tentatively.
- Final Exam (40%) – time and place: TBA

Assignment evaluations

- Submit every homework on the blackboard. It is not acceptable to sent to TAs/instructors by emails or other ways.
- For each homework, some selected number of questions will be marked and graded.
- Please note it is important to submit your assignments on time! Late submission may resulting a deduction of points or even zero points. Please check your submission carefully after the submission in the blackboard to make sure the submission is successful and complete.

Participation of the lectures are essentially important, which will ultimately lead to better academic outcomes.